

Abhishek Singh

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Profile

Final-year IIT Patna undergraduate with hands-on experience in Python, deep learning, computer vision and AWS EC2 deployment. Built video anomaly detection pipelines, prototyped LLM-based explanations, and developed modular simulation software. Seeking an IT Engineer role in cloud applications and AI development.

Education

Indian Institute of Technology Patna | B.Tech, Electronics and Communication Engineering

2023–2027 (expected) | CPI: 7.00/10

Relevant coursework (completed / ongoing): Introduction to AI–ML; Programming and Data Structures; Random Signals and Stochastic Processes; Digital Signal Processing; Computer Organization and Architecture.

Technical Skills

Programming: Python, C++, C, SQL | Web: React, Vite

AI and data: PyTorch, TensorFlow, NumPy, Pandas, OpenCV; CNNs, autoencoders, anomaly detection

Cloud and tools: AWS EC2, Linux, Nginx, Cloudflare, Git | LLM integration: Groq API, Llama 3

Experience

Video Surveillance Research Intern | IIT Patna | May–July 2026

Under Prof. Mahesh Kumar Kolekar

- Developed a PyTorch 3D convolutional autoencoder pipeline for video anomaly detection, covering preprocessing, training, inference and evaluation.
- Designed Hybrid Patch Feature Scoring using reconstruction and latent-feature errors; achieved ROC-AUC of 0.734 on UCSD Ped2 versus a baseline of approximately 0.66–0.675.
- Prototyped gradient-based anomaly localization and Llama 3 explanations through the Groq API to make detected events easier to interpret.

Software Head | Inditronix AI Labs | April 2025–May 2026

- Led software architecture and coordinated a multidisciplinary team working on healthcare AI solutions; deployed infrastructure on AWS EC2 and developed ESP32-based IoT applications.

Selected Projects

Inditronix Website Deployment | AWS EC2, React, Vite, Nginx | September 2026

- Deployed a React/Vite application on Ubuntu EC2 with Nginx single-page application routing and Cloudflare Full TLS configuration.
- Managed a deployment update using a separate release directory, configuration validation and Nginx reload; retained the previous application directory for rollback.

Prabha / PEMAAN | Python, Numerical Modeling | IIT Patna | May 2026–Present

- Developed modular continuous-wave laser and signed-weight photonic multiply–accumulate models; validated power, wavelength, noise, sampling-rate and random-seed behavior.
- Ran 1,000 stochastic realizations to recover a modeled 1 MHz laser linewidth within approximately 5.1%; verified ideal MAC outputs across three optical power levels.

USAS Ingestor Engine | Video Data Pipeline

- Built a multi-camera CCTV and DOT stream ingestion pipeline with synchronization, preprocessing and storage modules for surveillance analytics.

Achievements and Activities

- GATE ECE 2026: AIR 1046, score 662. Codeforces: Pupil (peak 1241). CodeChef: 3-star (peak 1643).
- Shortlisted for Amazon ML Summer School 2026; participated in Kaggle healthcare, loan-payback and rainfall prediction competitions.

Creator, IITaku – Gyan, Code and Anime: YouTube channel with 8,000+ subscribers.